

Coupled Graph–Policy Distillation for Personalized Medication Safety in Older Adults with Multimorbidity

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Abstract—Large language model (LLM) agents can support medication review between clinical visits, but safe choices for older adults with multimorbidity depend on conditions, medications, and geriatric risks that users may omit. We introduce ATLAS, a coupled graph–policy distillation framework for patient-adaptive medication safety. ATLAS structures guideline evidence as a medication-safety graph. Targeted questions update the patient state and distill relevant relations into a patient-specific medication conflict graph (PMCG). A risk-first multi-agent policy uses the PMCG to screen contraindications, assess cautions and monitoring needs, identify safer alternatives, and verify the final medication plan. We also introduce GeriMedBench, an interactive benchmark that tests safety-critical information acquisition and evidence-based decision revision. Across a European non-interactive multimorbidity benchmark, an Asian interactive multimorbidity benchmark, and an Asian non-interactive cross-guideline benchmark, ATLAS achieves the strongest complete-decision performance among the compared systems. On the European non-interactive multimorbidity benchmark, it exceeds the strongest proprietary LLM baseline by 53.73 points in Strict Success Rate and 14.63 points in overall safety reasoning score (OSRS), with no unsafe recommendations under the automated evaluator. A blinded clinician evaluation gives ATLAS higher mean ratings across all five criteria and flags potentially unsafe recommendations in one ATLAS case and two Gemini cases.

Index Terms—medication safety, multimorbidity, large language model agents, multi-agent systems, knowledge graphs, geriatric care

I. INTRODUCTION

Medication safety for older adults with multimorbidity depends on clinical facts that an initial query may omit. A user may describe a symptom or treatment goal without reporting a condition, medication, or geriatric risk that changes the safety of a candidate treatment. An agent that treats the first message as a complete patient profile can produce a plausible but unsafe recommendation. Figure 1 illustrates this gap. Safe medication decision support must identify decision-changing information, ask focused questions, and revise the medication plan as new evidence emerges. ATLAS supports clinician and pharmacist judgment rather than replacing it [1].

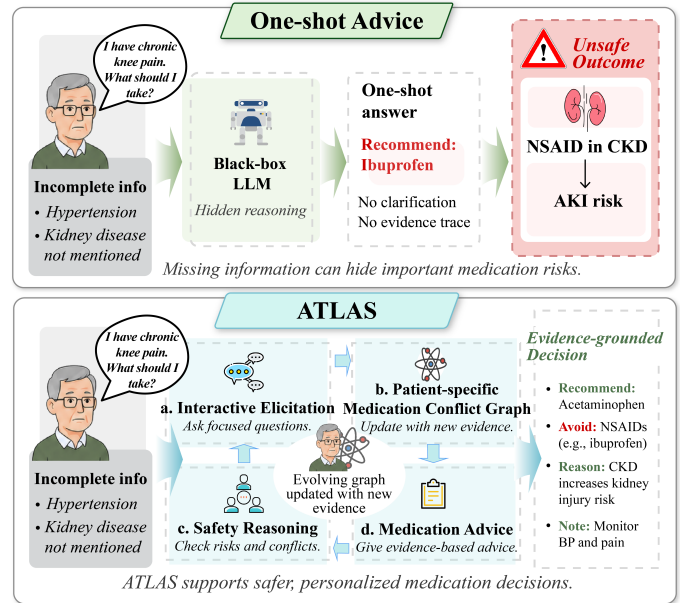


Fig. 1. One-shot medication advice versus ATLAS. A black-box LLM may miss unreported risks, while ATLAS elicits missing information, updates the patient-specific medication conflict graph (PMCG), and supports an evidence-grounded decision through safety-first reasoning.

This task poses two linked challenges. Guideline knowledge spans many medications, conditions, risks, monitoring requirements, and alternatives, while each patient requires a small and changing subset. The system must also convert this evidence into an ordered decision process that prioritizes avoidance, cautions, alternatives, and verification.

We propose **ATLAS**, a coupled graph–policy distillation framework. ATLAS builds a provisional patient-specific medication conflict graph (PMCG), uses unresolved graph relations to select questions, and updates the PMCG with each response. A symbolic risk-first policy then guides conflict screening, caution assessment, alternative selection, targeted revision, and final verification. The graph identifies which evidence

matters. The policy determines how that evidence changes the medication plan.

We introduce **GeriMedBench** to evaluate whether an agent can acquire missing safety evidence under a query budget and use that evidence to revise its decision. We evaluate ATLAS on a European non-interactive multimorbidity benchmark, an Asian interactive multimorbidity benchmark, and an Asian non-interactive cross-guideline benchmark. On the European non-interactive multimorbidity benchmark, ATLAS exceeds the strongest proprietary LLM baseline by 53.73 points in Strict Success Rate and 14.63 points in overall safety reasoning score (OSRS), with no unsafe recommendations. A blinded clinician evaluation gives ATLAS higher mean ratings across all five criteria. Our contributions are fourfold:

- 1) *Coupled Graph–Policy Distillation.* We distill patient-relevant guideline relations into a PMCG and teacher trajectories into a symbolic risk-first policy executed by a structured multi-agent system.
- 2) *Graph-Guided Interactive Personalization.* Unresolved graph dependencies trigger targeted questions, and each response updates the PMCG and selectively revises the safety assessments and medication decisions it supports.
- 3) *GeriMedBench.* We introduce an interactive benchmark for evaluating safety-critical information acquisition, budgeted questioning, and evidence-grounded, trace-consistent medication decisions.
- 4) *Cross-Setting and Cross-Protocol Evaluation.* We evaluate ATLAS on a European non-interactive multimorbidity benchmark, an Asian interactive multimorbidity benchmark, and an Asian non-interactive cross-guideline benchmark.

II. RELATED WORK

Drug Recommendation. Drug recommendation methods combine molecular structures, DDI knowledge, and patient records. SafeDrug [2] models DDIs with molecular graphs, while MoleRec [3] captures patient-specific effects of drug substructures. CT-PASMR [4] and EDRMM [5] improve drug representations through hybrid and multi-granularity modeling, while TraceDR [6], DrugRec [7], and Hyper-Geri [8] incorporate knowledge graphs, traceability, external knowledge, or hypergraph learning. These methods assume fixed patient profiles. They do not ask for missing multimorbidity information or revise medication-safety decisions through interaction.

LLM-Based Clinical Agents. LLMs increasingly support clinical decision-making through multi-agent collaboration. MedAgents [9] uses multidisciplinary discussion for medical question answering, ClinicalAgent [10] targets clinical trial tasks, DrAgent [11] combines clinical tools with recursive learning, and MedRad [12] integrates knowledge engineering with retrieval-augmented reasoning. Related work also explores MDAgents [13]. These systems address clinical reasoning and agent collaboration, but they do not center on patient-adaptive medication safety for older adults with multimorbidity.

Process-Oriented Agent Evaluation. Agent evaluation has expanded from final-answer accuracy to interactive reasoning and safety. AgentBench [14] evaluates agents across interactive environments, and MedicalAgentsBench [15] focuses on multi-step clinical reasoning and treatment planning. NO-HARM [16], MATRIX [17], CSEDB [18], and CARES [19] assess medical harm, hazardous clinical dialogues, safety and efficacy, or adversarial robustness. However, existing benchmarks do not specifically test whether medication agents can identify missing safety-critical information, ask informative questions under a limited budget, and use newly acquired evidence to revise decisions consistently.

III. METHOD

ATLAS treats medication decision support as an evidence-based consultation. It updates a structured patient state, distills relevant guideline relations into a PMCG, and executes a risk-first policy over the graph. The PMCG identifies unresolved safety dependencies. The policy selects questions, revises affected decisions, and verifies the final medication plan. Figure 2 summarizes graph distillation and risk-first policy execution.

A. Patient State and Three-Layer Agent Architecture

At each dialogue round t , ATLAS represents the patient state with five components:

$$s_t = (D_t, M_t, A_t, R_t, C_t), \quad (1)$$

Here, D_t , M_t , A_t , R_t , and C_t represent the diagnosed conditions in the multimorbidity profile, current medications, age and geriatric factors, safety modifiers, and therapeutic context, respectively. The therapeutic context includes the presenting problem, treatment goal, and decision setting. ATLAS treats unreported information as unknown rather than assuming its absence.

ATLAS groups its agents into three layers. The *Orchestration and Context Layer* contains the Unified Orchestrator and Clinical State Grounder. The *Graph Personalization and Safety Audit Layer* contains the Drug Conflict Auditor and Geriatric Risk Auditor. The *Decision Synthesis and Verification Layer* contains the Alternative Agent, Revision Agent, Trace Verifier, and Safety Gate. A shared blackboard passes patient evidence, PMCG updates, and decision revisions across the layers.

B. Stage I: Clinical Intake and Treatment Goal Setting

At the start of each consultation, ATLAS identifies the primary clinical concern and treatment goal, then records the available patient-state components. In later rounds, the Clinical State Grounder converts each patient response into a structured state update and retains the original wording for explanation.

ATLAS does not use a fixed questionnaire. It builds a provisional PMCG and asks only about missing information that could change the medication decision.

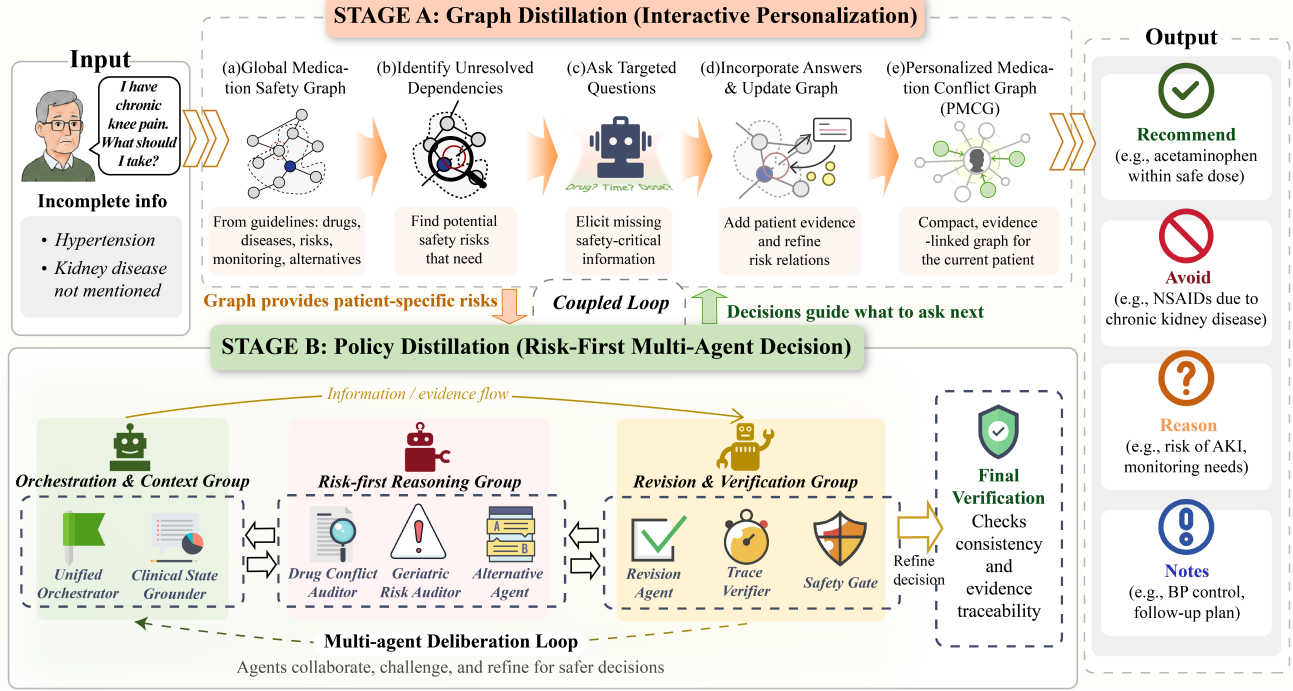


Fig. 2. ATLAS architecture. The graph branch updates an evolving PMCG through targeted questioning. The policy branch executes the distilled risk-first policy to assess conflicts, risks, alternatives, and recommendations, then verifies the final medication plan. The three agent groups correspond to the three functional layers defined in the text. The output panel illustrates an example report rather than the full evaluation schema, which contains recommendation, avoidance, caution, and alternative components.

C. Stage II: Progressive PMCG Distillation with Targeted Questioning

The guideline graph links medications, conditions, risk factors, therapeutic goals, monitoring requirements, alternatives, and supporting evidence [20]. ATLAS retains relations that match the current patient state, removes inapplicable relations, and marks those that depend on missing information as unresolved. Active relations support the current decision, while unresolved relations identify missing information that could still change the medication plan.

At each round, ATLAS ranks candidate questions first by unresolved risk severity and then by the number of medication decisions each question may affect. It uses the question history to break ties. After the patient responds, ATLAS adds the new evidence to the patient state and rebuilds the PMCG from the updated state and the relevant guideline graph:

$$\begin{aligned} s_{t+1} &= s_t \oplus \Phi(q_t, a_t), \\ G_{t+1}^p &= \mathcal{P}(G_t^g; g, s_{t+1}). \end{aligned} \quad (2)$$

Here, Φ maps question q_t and answer a_t to a structured state update, while $\oplus$ integrates the new evidence into the current patient state. G^g denotes the guideline graph, g the treatment goal, and $\mathcal{P}$ the personalization operator that constructs the updated PMCG G_{t+1}^p .

After each PMCG update, the Drug Conflict Auditor reassesses contraindications and medication-condition conflicts, while the Geriatric Risk Auditor reassesses age-related risks,

caution requirements, and monitoring needs. These audits determine whether the new evidence reveals a contraindication, resolves an earlier concern, introduces a monitoring requirement, or changes the preferred alternative. ATLAS stops questioning when no unresolved issue can meaningfully change the decision, no informative question remains, or the query budget runs out. By reconstructing the PMCG from the current patient state and relevant guideline evidence, ATLAS can apply the same procedure across multimorbidity settings and guideline sources. This coupling works in both directions. Unresolved PMCG relations guide the next question, while each response updates the graph and limits revisions to medication decisions affected by the changed relations.

D. Stage III: Consultation-Style Safety-First Decision

The current PMCG guides ATLAS to screen contraindications and serious conflicts before cautions, monitoring needs, and alternatives. ATLAS gives avoidance priority over treatment preference and removes high-risk options from the recommendation component. The Alternative Agent searches guideline evidence when an auditor excludes an option.

ATLAS uses its consultation agents as a multi-agent teacher. The teacher runs the 39 development cases under $K \in \{1, 2, 3\}$ and produces 117 trajectories. Each trajectory records questions, patient-state updates, PMCG transitions, risk assessments, revisions, stopping decisions, and evidence paths. Policy distillation extracts recurring guideline-consistent transitions and encodes them in a versioned YAML rule table.

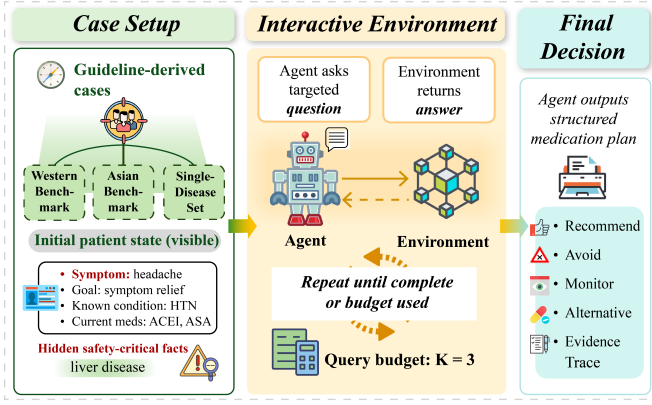


Fig. 3. GeriMedBench, an interactive medication-safety benchmark. The case-setup panel summarizes the evaluation sets, while only the Asian Multimorbidity Evaluation Set uses the interactive protocol. Under this protocol, an agent queries an incomplete patient state with hidden safety facts and revises its final decision under a budget of $K = 3$.

The symbolic policy governs question selection, risk priority, reconciliation, revision, stopping, and verification. It contains no case identifiers, evaluation labels, or learned parameters.

During inference, the agents execute this policy over the PMCG. The Clinical State Grounder applies alias normalization, entity canonicalization, negation handling, uncertainty handling, and component-aware parsing. PMCG-linked templates generate questions and evidence-grounded explanations. The Revision Agent updates decisions and evidence paths that depend on changed relations. ATLAS invokes no external LLM during inference. The supplementary material reports the policy, trajectory schema, teacher configuration, and inference procedure.

E. Stage IV: Medication Reconciliation and Decision Verification

The Revision Agent resolves overlaps among the recommendation, avoidance, and caution components using the priority order of avoidance, caution, and recommendation. It then checks each alternative against the revised medication components. The Trace Verifier links each claim to the current PMCG and evidence path. The Safety Gate checks component consistency and unresolved safety conflicts. A failed check returns the affected decision to the Revision Agent. ATLAS releases the medication plan after both checks pass.

F. GeriMedBench: Interactive Medication Safety Benchmark

GeriMedBench frames medication safety as an interactive task rather than a decision based on a complete patient profile. Each case provides an initial public state, hidden safety-critical facts that may change the decision, a response environment, and a guideline-grounded reference. As Figure 3 shows, the agent interacts with the environment under a limited query budget before producing a structured medication-safety decision [21].

At round t , the system observes the public state s_t and either asks a question q_t or returns a medication decision y_t .

When the system asks a question, the environment returns a patient answer a_t based on the hidden patient state. The system then updates the public state using only the newly revealed evidence:

$$s_{t+1} = s_t \oplus \phi(q_t, a_t). \quad (3)$$

Here, ϕ extracts the patient information that a_t reveals, while $\oplus$ adds this information to the public state without exposing any other hidden facts. The query budget K sets the maximum number of questions the system can ask.

The interaction ends when the system returns a final decision or exhausts its budget of K questions. Static benchmarks test whether a system can reason from a complete patient profile. GeriMedBench instead tests whether a system can identify decision-relevant missing information, acquire it efficiently, and use the new evidence to revise its final decision. The evaluation measures patient-state and decision updates, final medication safety, and consistency with the evidence trace. Clinical guidelines ground all patient facts, safety relations, and reference decisions.

IV. EXPERIMENTS AND RESULTS

We evaluate ATLAS on a European non-interactive multimorbidity benchmark, an Asian interactive multimorbidity benchmark, and an Asian non-interactive cross-guideline benchmark. The experiments measure complete medication decisions, safety, evidence acquisition, targeted revision, component contributions, and cross-guideline robustness.

A. Experimental Setup

Datasets and protocols. We use a separate 39-case development set for policy construction and system calibration. No policy or inference component accesses the evaluation references. Only the frozen offline evaluator reads the test labels.

The Western Multimorbidity Evaluation Set ($N = 201$) serves as the non-interactive multimorbidity benchmark. It is grounded in the AGS Beers Criteria [22], STOPP/START version 3 [23], the FORTA 2024 list [24], EURO-FORTA version 2 [25], and the deprescribing guideline [26].

The Asian Multimorbidity Evaluation Set ($N = 76$) serves as the interactive multimorbidity benchmark under GeriMedBench, where each agent may ask up to $K = 3$ questions. It is grounded in the Korean Delphi criteria [27], the Korean consensus criteria [28], and the Japanese guidance on appropriate medication use in older adults [29].

The Cross-Regional Single-Disease Generalization Set ($N = 612$) serves as the non-interactive cross-guideline benchmark with candidate-constrained medication selection. Its reference decisions are grounded in the ADA guideline for diabetes [30], the AHA/ACC/HFSA guideline for heart failure [31], and the KDIGO guideline for chronic kidney disease [32].

Clinical guidelines define patient facts, medication relations, and reference decisions. Each multimorbidity case contains a supported recommendation and guideline-grounded avoidance,

Method	Strict Success	M_{rec} F1	M_{avoid} Recall	M_{caution} F1	M_{alt} F1	Unsafe Rate	OSRS
<i>Foundation Models</i>							
Mistral-Small-3.2-24B	2.49	78.44	87.56	42.29	9.95	12.44	63.36
Qwen3-30B-A3B	16.42	26.37	98.01	62.19	65.17	1.99	70.85
DeepSeek-R1-Distill-Qwen-32B	26.87	85.41	95.52	43.78	53.73	3.48	74.58
MedGemma-27B-Text	1.99	60.36	92.54	57.21	6.47	7.46	65.37
GPT-5	32.34	69.15	99.00	41.29	49.12	0.50	71.37
Claude Opus 4.6	34.83	88.56	98.01	46.77	85.32	0.00	82.30
Gemini 3.1 Pro Preview	38.31	63.68	99.00	70.65	80.60	0.00	82.58
<i>Knowledge-Augmented Models</i>							
Llama-3.3-70B-Instruct	0.00	79.60	96.52	51.74	1.00	0.00	68.82
RotatE	0.00	30.85	13.43	20.40	0.00	62.69	28.26
<i>Multi-Agent Systems</i>							
MDAgents	6.97	32.34	93.53	79.60	7.13	0.00	69.42
<i>Ours</i>							
ATLAS	92.04	92.50	100.00	75.47	92.84	0.00	97.21
Δ vs. Best	$\uparrow 53.73$	$\uparrow 3.94$	$\uparrow 1.00$	$\downarrow 4.13$	$\uparrow 7.52$	0.00	$\uparrow 14.63$

TABLE I

MAIN RESULTS ON THE WESTERN MULTIMORBIDITY EVALUATION SET ($N = 201$). VALUES ARE PERCENTAGES EXCEPT OSRS. BEST VALUES ARE BOLD. ARROWS IN THE Δ ROW INDICATE THE DIRECTION OF CHANGE RELATIVE TO THE BEST NON-ATLAS RESULT.

Method	Revision Acc. $\uparrow$	Final Strict $\uparrow$	Trace Consistency $\uparrow$	Unsafe Rate $\downarrow$	Agent OSRS $\uparrow$
<i>Foundation Models</i>					
Qwen3-30B-A3B	43.69	3.95	75.00	10.53	55.03
DeepSeek-R1-Distill-Qwen-32B	38.64	2.63	65.79	13.16	49.03
MedGemma-27B-Text	35.43	0.00	81.58	0.00	53.63
GPT-5	40.60	2.63	81.58	0.00	58.98
Claude Opus 4.6	40.51	5.26	81.58	0.00	62.87
Gemini 3.1 Pro Preview	43.18	7.89	73.68	2.63	60.54
<i>Knowledge-Augmented Models</i>					
Llama-3.3-70B-Instruct	42.11	6.58	84.21	3.95	58.63
<i>Ours</i>					
ATLAS	44.17	23.68	85.53	0.00	64.09
Δ vs. Best	$\uparrow 0.48$	$\uparrow 15.79$	$\uparrow 1.32$	0.00	$\uparrow 1.22$

TABLE II

GERIMEDBENCH RESULTS ON THE ASIAN MULTIMORBIDITY EVALUATION SET ($N = 76$, $K = 3$). VALUES ARE PERCENTAGES EXCEPT AGENT OSRS. BEST VALUES ARE BOLD. THE FINAL ROW REPORTS DIFFERENCES BETWEEN ATLAS AND THE BEST NON-ATLAS RESULT IN EACH COLUMN.

Method	Correct.	Safety	Complete.	Action.	Evidence	Unsafe
Gemini 3.1 Pro Preview	3.77	4.12	3.67	3.85	3.80	2/40
ATLAS	4.11	4.35	4.01	4.01	4.08	1/40

TABLE III

BLINDED CLINICIAN EVALUATION ON 40 BENCHMARK-STRATIFIED CASES. SCORES USE A FIVE-POINT SCALE AND AVERAGE ALL REVIEWER-CASE RATINGS. UNSAFE CASES USE CASE-LEVEL MAJORITY JUDGMENTS.

caution, monitoring, and alternative requirements. These relations define M_{rec} , M_{avoid} , M_{caution} , and M_{alt} . LLMs diversify scenario wording but do not create safety labels.

Compared methods and evaluation.: We compare ATLAS with Mistral-Small-3.2-24B, Qwen3-30B-A3B, DeepSeek-R1-Distill-Qwen-32B, MedGemma-27B-Text, GPT-5, Claude Opus 4.6, and Gemini 3.1 Pro Preview. Knowledge-augmented baselines include Llama-3.3-70B-Instruct with BM25 retrieval and RotatE. MDAgents provides the multi-agent baseline.

All LLM baselines receive the same visible inputs, candidate set, structured-output instruction, parser, and failure rules. We do not tune prompts on evaluation cases. No inference method accesses test references. The supplementary material reports prompts, model snapshots, decoding settings, retrieval settings, retries, and invalid-output handling. These comparisons measure end-to-end systems and do not isolate guideline access from graph personalization and multi-agent control.

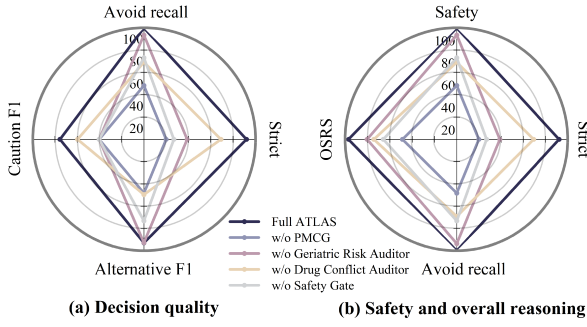


Fig. 4. Core safety-reasoning ablations. Panel (a) compares decision-quality metrics, whereas Panel (b) summarizes safety and overall performance. Full ATLAS is compared with four component ablations. Safety is defined as $100 - \text{Unsafe Recommendation Rate}$.

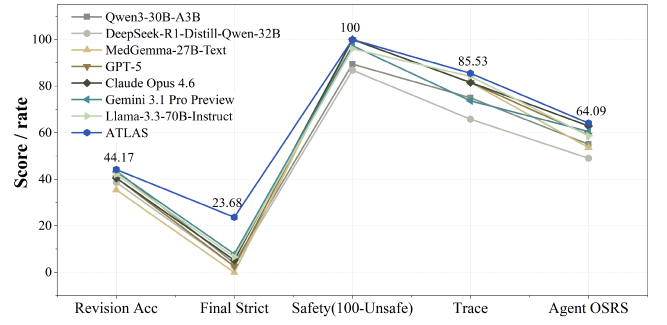


Fig. 5. GeriMedBench performance on the Asian Multimorbidity Evaluation Set ($N = 76$, $K = 3$). Safety is defined as $100 - \text{Unsafe Rate}$. ATLAS is highlighted in blue. Rates are reported as percentages, while Agent OSRS is reported on a 0–100 scale.

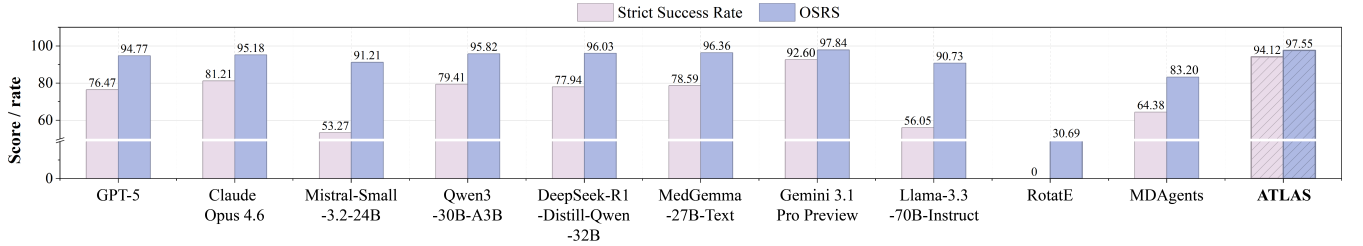


Fig. 6. Single-disease comparison on the Cross-Regional Single-Disease Generalization Set ($N = 612$), showing Strict Success Rate and OSRS across methods. Rates are reported as percentages, while OSRS is reported on a 0–100 scale.

Blinded clinician evaluation.: Three clinicians independently reviewed 40 benchmark-stratified cases, with 20 cases from the Western evaluation and 20 from GeriMedBench. Each reviewer compared one ATLAS output with one Gemini 3.1 Pro Preview output. We randomized case order and A/B assignment for each reviewer. We removed model identifiers and automated scores and presented both outputs through the same structured template. Reviewers score clinical correctness, medication safety, decision completeness, actionability, and evidence consistency on a five-point scale [33]. They also mark potentially unsafe recommendations and select the preferred output. We report mean scores, case-level majority judgments, and Krippendorff’s alpha.

Metrics.: The four medication components are recommendation, avoidance, caution, and alternative selection. For static evaluation, we report Strict Success, M_{rec} F1, M_{avoid} Recall, M_{caution} F1, M_{alt} F1, Unsafe Recommendation Rate, Trace Pass Rate, and OSRS. We use recall for M_{avoid} because a false negative omits a medication that the patient should avoid. F1 penalizes omissions and unsupported additions in the other components. For GeriMedBench, we report Revision Accuracy, Final Strict, Unsafe Recommendation Rate, Trace Consistency, and Agent OSRS. The supplementary material reports Information Gain and Query Efficiency.

$$\text{Strict Success Rate} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \begin{bmatrix} C_{\text{rec},i} = C_{\text{avoid},i} = 1, \\ C_{\text{caution},i} = C_{\text{alt},i} = 1, \\ C_{\text{unsafe},i} = C_{\text{trace},i} = 1 \end{bmatrix}. \quad (4)$$

For case i , $C_{s,i} = 1$ when component s is correct. $C_{\text{unsafe},i} = 1$ means that the output contains no unsafe recommendation. Strict Success requires correctness across all six components. Component metrics separate individual decision errors from failure of the complete output.

$$\text{OSRS} = 0.25R_{\text{avoid}} + 0.20R_{\text{rec}} + 0.15R_{\text{caution}} + 0.15R_{\text{alt}} + 0.15T + 0.10(1 - U). \quad (5)$$

OSRS uses component recall, Trace Pass Rate T , and Unsafe Recommendation Rate U . We give avoidance recall the largest weight because missed avoidance relations carry direct safety risk. We normalize each rate to $[0, 1]$ and report OSRS on a 0–100 scale. We fixed the weights before final evaluation.

B. Results on the Western Multimorbidity Evaluation Set

Table I reports results on the European non-interactive multimorbidity benchmark. ATLAS achieves 92.04% Strict Success and an OSRS of 97.21. It exceeds the best non-ATLAS results by 53.73 points in Strict Success and 14.63 points in OSRS, with no unsafe recommendations under the automated evaluator. Strict Success measures joint correctness rather than an equal gain in every decision component. Several baselines remain competitive on individual components, but ATLAS completes the full structured decision more often.

Figure 4 shows that each safety component protects a different part of the decision. Removing PMCG personalization reduces Strict Success from 92.04% to 19.90%, lowers OSRS

from 97.21 to 48.78, and raises Unsafe Rate to 51.74%. Removing the Geriatric Risk Auditor lowers M_{caution} F1 to 40.30%. Removing the Drug Conflict Auditor lowers M_{avoid} Recall to 69.15% and raises Unsafe Rate to 30.85%. Removing the Safety Gate reduces Strict Success from 92.04% to 26.87% and increases the Unsafe Recommendation Rate from 0% to 26.87%.

C. GeriMedBench on the Asian Multimorbidity Evaluation Set

Table II reports the detailed quantitative results, while Figure 5 summarizes the corresponding performance profiles on the Asian interactive multimorbidity benchmark. ATLAS achieves the highest Revision Accuracy of 44.17%, Final Strict of 23.68%, Trace Consistency of 85.53%, and Agent OSRS of 64.09, with no unsafe recommendations under the automated evaluator. The Final Strict and Revision Accuracy scores expose the remaining challenges in information acquisition and decision revision. Although ATLAS outperforms the compared systems, these results leave substantial room for improvement.

D. Blinded clinician evaluation.

Table III complements the automated results with blinded clinician judgments. ATLAS receives higher mean scores than Gemini 3.1 Pro Preview for clinical correctness, medication safety, decision completeness, actionability, and evidence consistency. Case-level majority judgments prefer ATLAS in 28 cases, Gemini in 5 cases, and report 7 ties. Reviewers flag potentially unsafe recommendations in 1 ATLAS case and 2 Gemini cases. Ordinal Krippendorff’s α ranges from 0.33 to 0.71 across the five criteria. These results support the expert-rated clinical quality of ATLAS outputs within the reviewed sample. They do not establish real-world clinical effectiveness.

E. Cross-Regional Single-Disease Generalization Set

The Asian non-interactive cross-guideline benchmark tests medication-safety performance outside the multimorbidity settings. As shown in Figure 6, ATLAS achieves the highest Strict Success Rate of 94.12%, exceeding Gemini 3.1 Pro Preview by 1.52 percentage points. ATLAS also records an overall safety reasoning score of 97.55, within 0.29 points of Gemini’s best score of 97.84 and above all other methods. These results show that ATLAS delivers the strongest complete-decision performance while maintaining strong overall safety reasoning across guideline sources.

F. Error Analysis and Limitations

We group failures into missed information, patient-state and PMCG errors, risk-classification errors, unsuitable alternatives, revision errors, evidence-alignment errors, and schema errors. On GeriMedBench, we inspect question selection, state updates, PMCG transitions, revisions, and trace grounding. The low Final Strict score exposes errors across the interactive pipeline.

Strict Success and OSRS capture different properties. Strict Success requires joint correctness. OSRS aggregates component recall, trace quality, and unsafe outcomes. On

the Asian non-interactive cross-guideline benchmark, ATLAS leads Strict Success, while Gemini 3.1 Pro Preview leads OSRS by 0.29 points. No single aggregate metric establishes clinical superiority.

Our comparisons evaluate complete systems and do not separate structured guideline access from graph personalization and multi-agent control. The benchmarks use guideline-derived cases, structured schemas, and candidate-constrained single-disease decisions. They do not test open-ended prescribing, patient communication, or clinical outcomes. The blinded expert study covers a small sample and does not establish clinical effectiveness. More broadly, benchmark performance alone is insufficient to establish readiness for clinical deployment [34]. Future work will extend expert review and prospective evaluation.

V. CONCLUSION

We present ATLAS, a coupled graph-policy distillation framework for medication safety in older adults with multimorbidity. ATLAS identifies missing safety information, updates a PMCG, and revises decisions that depend on new evidence. A symbolic risk-first policy guides conflict screening, caution assessment, alternative selection, and final verification.

We also introduce GeriMedBench to test safety-critical information acquisition and evidence-based revision. Across a European non-interactive multimorbidity benchmark, an Asian interactive multimorbidity benchmark, and an Asian non-interactive cross-guideline benchmark, ATLAS achieves the strongest complete-decision performance among the compared systems and records no unsafe recommendations under the automated evaluator. A small blinded clinician evaluation gave ATLAS higher mean ratings across all five criteria and fewer majority-flagged unsafe cases than Gemini. These results support graph-guided questioning and evidence-based revision across the tested settings. They do not establish real-world clinical effectiveness.

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APPENDIX

- A. Claim–Evidence Map
- B. Benchmark Composition and Data Protocols
- C. ATLAS Implementation Details
- D. GeriMedBench Environment and Evaluation
- E. Baseline Configuration and Output Handling
- F. Additional Quantitative Results
- G. Qualitative Case Study and Error Taxonomy
- H. Blinded Clinician Evaluation
- I. Shared Prompt Templates and Output Schema

Table A1 gives a direct route from each main-paper claim to the corresponding protocol, artifact, or analysis. It also marks the boundary of each claim, so evidence from one setting is not treated as evidence for another.

This section separates dataset construction, information visibility, and output supervision. Table B1 identifies the role of each evaluation set, while Tables B2 and B3 specify what the evaluated systems can observe and what they must return.

B.1 Evaluation Sets

The four sets are not interchangeable: each isolates a different stage of development or evaluation. Table B1 therefore reports both protocol and primary purpose rather than presenting the sets as a single pooled benchmark.

The evaluation separates development, non-interactive multimorbidity reasoning, interactive multimorbidity reasoning, and cross-guideline single-disease generalization. Table B1 summarizes the distinct role of each split.

B.2 Case Construction and Label Provenance

The construction procedure preserves a strict separation between language variation and clinical supervision. This distinction is important when interpreting the benchmark inventory and information boundaries in Tables B1 and B2.

The evaluation sets contain guideline-derived, constructed clinical cases rather than identifiable patient records. Guideline sources define patient facts, medication relations, and reference decisions. Each multimorbidity case contains a supported recommendation and guideline-grounded avoidance, caution, monitoring, and alternative requirements. Language models vary surface wording only; they do not create safety labels.

This design supports controlled comparison of medication-safety reasoning over structured guideline evidence. It does not estimate population prevalence, treatment effect, or real-world clinical outcome.

B.3 Visible State, Hidden State, and Output Schema

Table B2 contrasts the static and interactive information boundaries at inference time. Table B3 then decomposes the required output so that an apparently plausible recommendation cannot compensate for a missed avoidance, caution, alternative, unsafe indicator, or evidence trace.

This section makes the implementation boundary explicit from patient-state representation through frozen-policy execution. Tables C1–C4 document the graph relations, agent responsibilities, teacher trajectories, and freeze checks that support the reported ATLAS configuration.

C.1 Patient State and PMCG Representation

Table C1 summarizes the PMCG relation types and shows how each relation affects a medication decision. The separation between support, conflict, caution, alternative, evidence, and unresolved-dependency relations allows updates to remain localized to the affected claims.

At dialogue round t , ATLAS represents the patient state as $s_t = (D_t, M_t, A_t, R_t, C_t)$, where the components encode diagnosed conditions, current medications, age and geriatric factors, safety modifiers, and therapeutic context. The system retains unknown values as unknown rather than converting missing information into negative evidence.

C.2 Agent Responsibilities

Table C2 maps each functional layer to its agents and responsibilities. The decomposition also clarifies which components can revise the patient state, personalize the graph, audit risk, or reconcile a final decision.

C.3 Question Ranking, Revision, and Stopping

The ordered procedure below links question selection to unresolved PMCG relations rather than to a fixed questionnaire. Its stopping rule prevents additional questions once the remaining uncertainty cannot change a medication component.

ATLAS ranks candidate questions by unresolved risk severity and then by the number of medication decisions that the answer may affect. It uses the question history to break ties. Each answer updates the structured patient state, rebuilds the relevant PMCG subgraph, and triggers a targeted re-audit of the affected decisions.

- 1) Build a provisional PMCG from the current patient state and relevant guideline graph.
- 2) Identify unresolved relations whose resolution may change recommendation, avoidance, caution, or alternative decisions.
- 3) Rank questions by risk severity, decision coverage, and question history.
- 4) Integrate the revealed evidence and rebuild the personalized graph.
- 5) Re-run the affected safety audits and revise only dependent decision components.
- 6) Stop when no unresolved issue can change the decision, no informative question remains, or the query budget is exhausted.

C.4 Teacher Trajectories and Policy Distillation

Table C3 lists every field retained in a teacher trajectory. Together, these fields preserve the connection between a reviewed question, the resulting state and graph updates, the revised decision, and its evidence path.

ATLAS uses its structured consultation agents as a multi-agent teacher. The teacher processes the 39 development cases under three bounded review settings, $K \in \{1, 2, 3\}$, producing 117 structured trajectories. Each trajectory records internal safety questions, patient-state updates, PMCG transitions, risk assessments, component-level revisions, stopping decisions,

Main claim	Primary evidence	Supplementary location
ATLAS personalizes medication safety reasoning to incomplete patient states.	Patient-state schema, PMCG relations, agent responsibilities, and the illustrative revision trace.	Sections C and G
GeriMedBench evaluates information acquisition and decision revision under a limited query budget.	Environment state, interaction loop, stopping rule, and interactive metrics.	Section D
ATLAS improves complete-decision performance across the evaluated settings.	Expanded Western results, GeriMedBench acquisition metrics, and the tabular single-disease comparison.	Section F
Different safety modules protect different decision components.	Metric-specific ablation effects and component interpretation.	Section F.4
Clinicians rate ATLAS above Gemini in the reviewed sample.	Review protocol, overall scores, benchmark-stratified scores, and preference counts.	Section H
No inference method reads evaluation references.	Reference-access boundaries and shared evaluation controls.	Sections B.3 and E.2
Teacher trajectories are distilled into the symbolic policy used by ATLAS inference.	Trajectory schema, extraction procedure, YAML policy, compilation manifest, and behavioral-equivalence verification.	Sections C.4–C.6

TABLE A1
CLAIM–EVIDENCE MAP FOR THE SUPPLEMENTARY MATERIAL.

Set	N	Protocol	Primary purpose
Development Set	39	Development only	Policy construction and system calibration. No reported test score is computed on this split.
Western Multimorbidity Evaluation Set	201	Non-interactive	Complete medication-safety decisions from a visible multimorbidity profile.
Asian Multimorbidity Evaluation Set	76	Interactive, $K = 3$	Question selection, state update, targeted revision, final safety, and trace consistency under GeriMedBench.
Cross-Regional Single-Disease Generalization Set	612	Non-interactive, candidate-constrained	Cross-guideline generalization outside the multimorbidity settings.

TABLE B1
BENCHMARK INVENTORY AND EVALUATION ROLE.

Element	Non-interactive sets	GeriMedBench
Visible patient state	Complete benchmark input provided at inference.	Initial public state with selected safety-critical facts omitted.
Hidden safety facts	Not applicable.	Held by the response environment and revealed only through a relevant question.
Query action	Not available.	The agent may ask up to $K = 3$ targeted questions.
Candidate medications	Provided according to the benchmark protocol.	Provided in the public case and held fixed across methods.
Final output	Recommendation, avoidance, caution, alternative, unsafe indicator, and evidence trace.	The same structured medication decision after interaction.
Reference access	Frozen offline evaluator only.	Environment uses hidden state for answers; final references remain evaluator-only.

TABLE B2
INFORMATION BOUNDARY ACROSS STATIC AND INTERACTIVE EVALUATION.

and evidence paths. Policy distillation removes case-specific fields, merges recurring guideline-consistent transitions, resolves conflicting actions by risk priority, and stores the resulting symbolic policy in a versioned YAML rule table. The policy governs question selection, risk priority, medication reconciliation, targeted revision, stopping, and evidence verification. It contains no evaluation-case identifiers, test labels, or learned parameters. A deterministic compilation step converts the YAML policy into the frozen runtime representation consumed by ATLAS inference. The compiler preserves rule conditions, priorities, actions, and evidence requirements. All

reported ATLAS experiments use this compiled frozen policy.

C.5 Policy Compilation and Freeze Verification

Table C4 reports the release checks used to compare the symbolic source policy, compiled runtime representation, and frozen experimental artifact. These checks are intended to detect implementation drift rather than to add new clinical rules after development.

The distillation pipeline writes the symbolic policy to a versioned YAML file. A deterministic compiler converts this source policy into the frozen JSON representation read by the inference engine. The compilation process does not in-

Component	Required content	Primary failure mode
Recommendation (M_{rec})	Supported medication option and clinically appropriate use context.	Omission or unsupported addition.
Avoidance (M_{avoid})	Medication that should not be used under the case-specific safety evidence.	Missed unsafe option.
Caution (M_{caution})	Monitoring, dose, duration, or conditional-use requirement.	Missing or unsupported caution.
Alternative (M_{alt})	Safer or guideline-supported substitute when a candidate is excluded.	Missing or unsuitable alternative.
Unsafe indicator (U)	Whether the final plan contains an unsafe recommendation.	Unsafe recommendation remains in the final output.
Evidence trace	Link from each claim to patient evidence and guideline-grounded reasoning.	Unsupported or internally inconsistent trace.

TABLE B3
STRUCTURED OUTPUT COMPONENTS AND THEIR PRINCIPAL FAILURE MODES.

PMCG relation	Source node	Target node	Decision role
Recommendation support	Condition, goal, or therapeutic context	Medication candidate	Supports inclusion in M_{rec} when no higher-priority risk overrides it.
Avoidance conflict	Condition, medication, or safety modifier	Medication candidate	Excludes the candidate and activates M_{avoid} .
Caution or monitoring dependency	Age factor, organ function, treatment duration, or comorbidity	Medication candidate	Adds a condition, monitoring need, or dose/duration restriction.
Alternative relation	Excluded or unsuitable candidate	Safer substitute	Populates M_{alt} after risk screening.
Evidence relation	Guideline source or patient fact	Decision claim	Supports trace verification and final explanation.
Unresolved dependency	Missing patient fact	Potential decision change	Triggers a targeted question when the answer can change the plan.

TABLE C1
CORE PMCG RELATION TYPES USED FOR PERSONALIZATION AND VERIFICATION.

Layer	Agent	Responsibility
Orchestration and Context	Unified Orchestrator	Coordinates the consultation, maintains the blackboard, orders agent calls, and enforces the stopping rule.
Orchestration and Context	Clinical State Grounder	Normalizes aliases, canonicalizes entities, handles negation and uncertainty, and converts responses into structured state updates.
Graph Personalization and Safety Audit	Drug Conflict Auditor	Checks medication-condition conflicts and contraindications and updates avoidance decisions.
Graph Personalization and Safety Audit	Geriatric Risk Auditor	Checks age-related risk, organ-function dependencies, cautions, and monitoring needs.
Decision Synthesis and Verification	Alternative Agent	Searches guideline evidence for a safer substitute when an option is excluded.
Decision Synthesis and Verification	Revision Agent	Updates only the decision components affected by changed PMCG relations and resolves component overlaps.
Decision Synthesis and Verification	Trace Verifier	Links each final claim to the current PMCG and evidence path.
Decision Synthesis and Verification	Safety Gate	Rejects internally inconsistent plans or plans with unresolved safety conflicts.

TABLE C2
FUNCTIONAL DECOMPOSITION OF THE ATLAS AGENTS.

introduce new clinical rules or case-specific conditions. The release records the source-policy hash, compiled-policy hash, compiler version, rule count, and compilation command. We verify that the compiled artifact preserves every rule condition, priority, action, and evidence requirement. We also compare the compiled policy with the frozen policy used for the reported experiments through structural comparison and case-

level behavioral testing. The verification requires identical structured outputs for the frozen and compiled policies on the development cases and all released ATLAS evaluation inputs. A mismatch causes the verification script to fail.

C.6 Inference Procedure

Algorithm 1 presents the complete inference order from state initialization to verified output. The procedure makes

Trajectory field	Description
trajectory_id	Anonymous development trajectory identifier.
review_budget	Bound on teacher-review rounds used to generate the development trajectory.
initial_state	Structured state before review.
questions	Teacher-internal safety checks and decision-changing information needs.
state_updates	Normalized evidence added during review.
pmcg_transitions	Activated, resolved, or revised graph relations.
risk_assessments	Conflict, geriatric-risk, caution, and monitoring judgments.
decision_revisions	Component-level changes linked to graph updates.
stopping_decision	Reason for terminating review.
evidence_paths	Patient-fact and guideline support for the final claims.
final_decision	Structured recommendation, avoidance, caution, alternative, unsafe, and trace fields.

TABLE C3
TEACHER-TRAJECTORY SCHEMA USED FOR SYMBOLIC POLICY DISTILLATION.

Verification item	Required result
Development trajectories	117
Development cases	39
Review settings	$K \in \{1, 2, 3\}$
Case-specific conditions in policy	None
Test-label access	None
YAML-to-runtime compilation	Deterministic
Rule conditions and priorities	Exact match
Case-level outputs	Identical
Policy used for reported experiments	Compiled frozen policy

TABLE C4
POLICY-DISTILLATION AND FREEZE-VERIFICATION CHECKS.

explicit that graph updates precede localized revision and that the trace verifier operates on the reconciled medication sets.

This section describes the interactive environment independently of any particular agent. Table D1 defines the information boundary, Table D2 defines the evaluation dimensions, and Table D3 connects pipeline failures to their observable consequences.

D.1 Environment State and Interaction Loop

Table D1 identifies which environment objects are visible to the agent and which remain evaluator-only. Algorithm 2 complements this boundary by showing the action loop under the fixed question budget.

GeriMedBench separates the initial public state from hidden safety-critical facts. The environment returns only the answer to the agent’s current question. It does not expose unrelated hidden facts. The interaction ends when the agent returns a final decision or uses all $K = 3$ questions.

D.2 Interactive Metrics

Table D2 defines the interactive metrics at the level of observable behavior. Reading the dimensions together distinguishes acquiring useful evidence from merely producing a strong final answer without an evidence-consistent revision path.

D.3 Interactive Failure Taxonomy

Table D3 assigns each failure stage an observable error and a downstream consequence. This stage-wise view supports

error analysis without treating all unsuccessful consultations as equivalent.

The interactive pipeline can fail before or after a relevant fact is acquired. The error taxonomy separates these stages so that a low Final Strict score is not reduced to final-answer accuracy alone.

This section records the comparison boundary used by the offline evaluator. Table E1 groups the compared systems, and Table E2 lists the shared inputs, decoding controls, parsing rules, and reference-access restrictions.

E.1 Compared Systems

Table E1 organizes the baselines by system type so that foundation-model, retrieval-augmented, multi-agent, and proposed-system results can be interpreted consistently.

E.2 Shared Evaluation Settings

Table E2 records the controls shared within each comparison protocol. The common parser and failure handling are especially important because malformed output must not be confused with a clinically correct structured decision.

E.3 Structured Output Handling

The parser applies the schema introduced in Section I without consulting evaluation references. Consequently, schema validation and clinical scoring remain separate stages of the evaluation pipeline.

The parser extracts the first valid JSON object, normalizes aliases, and validates the required fields. A response that

Algorithm 1: ATLAS inference procedure.**Input:** Visible patient state s_0 , candidate set $\mathcal{C}$, guideline graph G , compiled frozen policy Π , and query budget K .

- 1: Normalize the input and initialize a provisional PMCG G_0^p .
- 2: **for** $t = 0, \dots, K$ **do**
- 3: Audit contraindications, geriatric risks, cautions, and alternatives.
- 4: Identify unresolved relations that can change the medication plan.
- 5: **if** no informative unresolved relation remains **then break**.
- 6: Select the highest-priority question q_t .
- 7: Receive answer a_t and update $s_{t+1} = s_t \oplus \Phi(q_t, a_t)$.
- 8: Rebuild the affected PMCG relations and revise dependent decisions.
- 9: **end for**
- 10: Reconcile avoidance, caution, recommendation, and alternative components.
- 11: Verify evidence paths and apply the Safety Gate.
- 12: **return** the structured medication-safety decision and trace.

Environment object	Contents	Access rule
Public case	Presenting problem, therapeutic goal, known conditions, current medications, candidates, and visible context.	Visible to the agent at the start.
Hidden state	Safety-critical facts that may alter avoidance, caution, monitoring, or alternative decisions.	Accessible only to the environment.
Response map	Question-to-answer mapping grounded in the hidden state.	Returns one answer for the current question.
Interaction trace	Questions, answers, state updates, and decision revisions.	Recorded for evaluation.
Reference decision	Guideline-grounded final medication components and trace requirements.	Frozen evaluator only.

TABLE D1

GERIMEDBENCH ENVIRONMENT OBJECTS AND ACCESS BOUNDARY.

Algorithm 2: Interactive evaluation loop used by GeriMedBench.**Input:** Public state s_0 , hidden state h , and query budget $K = 3$.

- 1: $t \leftarrow 0$.
- 2: **while** $t < K$ and the agent has not returned a final decision **do**
- 3: Agent selects either ASK(question) or FINAL(decision).
- 4: **if** the action is ASK **then**
- 5: Environment returns only answer(h , question).
- 6: Agent updates the public state with the revealed evidence.
- 7: **end if**
- 8: $t \leftarrow t + 1$.
- 9: **end while**
- 10: Evaluate the final decision, revision behavior, and evidence trace.

Metric	What it measures	Interpretation
Information Gain	Whether the selected questions reveal safety-relevant hidden information.	Higher values indicate that the interaction acquires more decision-relevant evidence.
Query Efficiency	Safety-relevant information obtained relative to the limited question budget.	Higher values indicate that fewer or more focused questions obtain useful evidence.
Revision Accuracy	Whether newly revealed evidence produces the required component-level decision change.	A high score requires both correct acquisition and correct downstream revision.
Final Strict	Joint correctness of the final structured decision after interaction.	A single missing or unsupported component causes case-level failure.
Trace Consistency	Consistency between the interaction trace, acquired facts, and final claims.	Penalizes unsupported claims and revisions disconnected from new evidence.
Agent OSRS	Overall safety reasoning across final components, unsafe outcomes, and trace quality.	Summarizes safety reasoning on a 0–100 scale.

TABLE D2

GERIMEDBENCH EVALUATION DIMENSIONS.

remains invalid after the shared runner policy receives the common failure treatment. The evaluator does not recover missing medication components from free text.

Fairness principle. The comparison fixes visible information, candidate medications, output schema, parser, and failure rules. It evaluates complete systems and does not isolate the

Failure stage	Observable error	Downstream consequence
Question selection	The agent asks about a low-impact fact while a high-risk dependency remains unresolved.	Hidden safety evidence remains unavailable.
State grounding	The response is misparsed, negated, or assigned to the wrong patient-state field.	The PMCG reflects an incorrect patient state.
PMCG update	A newly revealed fact does not activate or resolve the expected graph relation.	The relevant audit is not triggered.
Risk classification	The system identifies the fact but assigns the wrong avoidance or caution level.	Unsafe recommendation or incomplete monitoring.
Revision	The agent acquires the right fact but fails to update the affected decision component.	Low Revision Accuracy and Final Strict.
Evidence alignment	The final claim does not follow from the acquired evidence path.	Trace inconsistency.
Schema handling	The output is incomplete or cannot be parsed into the required structure.	Case-level failure under strict evaluation.

TABLE D3
ERROR TAXONOMY FOR THE INTERACTIVE PIPELINE.

Category	Methods
Foundation models	Mistral-Small-3.2-24B; Qwen3-30B-A3B; DeepSeek-R1-Distill-Qwen-32B; MedGemma-27B-Text; GPT-5; Claude Opus 4.6; Gemini 3.1 Pro Preview
Knowledge-augmented models	Llama-3.3-70B-Instruct with BM25 retrieval; RotatE
Multi-agent system	MDAgents
Proposed system	ATLAS

TABLE E1
METHOD CATEGORIES USED IN THE REPORTED COMPARISONS.

Setting	Value or rule
Visible input	Identical case fields and candidate set for all compared methods.
Output constraint	Shared structured JSON instruction and common target schema.
Temperature	0.0 for LLM-based baselines.
Maximum generation length	4096 tokens for LLM-based baselines.
Local serving context	vLLM <code>max_model_len</code> = 8192 for local model servers.
Retrieval	BM25 <code>top_k</code> = 8 for the retrieval-augmented baseline.
Parser	Shared robust JSON parser for LLM, RAG, and multi-agent outputs.
Evaluation references	Unavailable to all inference runners; readable only by the frozen offline evaluator.
Prompt tuning	No prompt tuning on evaluation cases.
Invalid output	Handled by the same runner-level retry and failure policy within each shared comparison protocol.

TABLE E2
SHARED BASELINE SETTINGS AND FAIRNESS CONTROLS.

independent contribution of guideline access, graph personalization, or multi-agent control.

This section provides numerical detail beyond the compact summaries in the main paper. Tables F1–F4 report expanded static results, interactive acquisition metrics, single-disease generalization, and component-level ablation effects.

F.1 Expanded Western Results for Proprietary Baselines and ATLAS

Table F1 expands the Western comparison with component-level scores, trace pass rate, unsafe rate, and OSRS. The additional columns show whether a method’s aggregate result reflects balanced medication reasoning or strength in only a subset of components.

Table F1 adds M_{avoid} F1 and Trace Pass Rate to the proprietary-model comparison. These metrics clarify why high avoidance recall alone does not guarantee a complete medication decision. ATLAS does not lead every component. Its advantage lies in joint completion: high recommendation quality, complete avoidance recall, stronger caution and alternative coverage, and no unsafe recommendation under the automated evaluator occur in the same structured output.

F.2 GeriMedBench Acquisition Metrics

Table F2 places acquisition and revision measures beside final decision quality. This joint presentation is necessary because interactive success depends on both obtaining decision-changing evidence and using it correctly.

Method	Strict Success	M_{rec} F1	M_{avoid} Recall	M_{avoid} F1	M_{caution} F1	M_{alt} F1	Unsafe	Trace	OSRS
GPT-5	32.34	69.15	99.00	90.05	41.29	49.12	0.50	100.00	71.37
Claude Opus 4.6	34.83	88.56	98.01	97.35	46.77	85.32	0.00	100.00	82.30
Gemini 3.1 Pro Preview	38.31	63.68	99.00	97.84	70.65	80.60	0.00	100.00	82.58
ATLAS	92.04	92.50	100.00	91.57	75.47	92.84	0.00	92.04	97.21

TABLE F1

EXPANDED RESULTS ON THE WESTERN MULTIMORBIDITY EVALUATION SET ($N = 201$). VALUES ARE PERCENTAGES EXCEPT OSRS.

Method	Info Gain	Query Efficiency	Revision Accuracy	Final Strict	Unsafe Rate	Trace Consistency	Agent OSRS
ATLAS	73.90	35.75	44.17	23.68	0.00	85.53	64.09

TABLE F2

ATLAS ACQUISITION AND FINAL-DECISION METRICS ON GERIMEDBENCH ($N = 76$, $K = 3$). VALUES ARE PERCENTAGES EXCEPT AGENT OSRS.

F.3 Tabular Single-Disease Comparison

Table F3 provides the numerical values underlying the main-paper visualization. Reporting the values directly makes the small differences in Strict Success and OSRS independently inspectable.

Under this evaluation, ATLAS leads Strict Success by 1.52 points over Gemini 3.1 Pro Preview, while Gemini leads OSRS by 0.29 points. The two metrics capture different properties, so the comparison supports stronger complete-decision performance rather than uniform superiority on every aggregate score.

F.4 Metric-Specific Ablation Effects

Table F4 relates each ablation to its primary observed effect. The comparison is component-specific because removing a module need not affect recommendation, avoidance, caution, alternative, trace, and unsafe outcomes equally.

This section follows individual evidence changes through the PMCG and final medication decision. Tables G1 and G2 provide concrete revision traces, while Table G3 supplies a common template for assigning observed failures to pipeline stages.

G.1 Illustrative Graph-Guided Revision

Table G1 traces the PMCG changes triggered by newly revealed evidence. Example G1 presents the corresponding structured medication sets and evidence paths after revision.

The following case study expands the knee-pain example in Figure 1 of the main paper. It is an illustrative trace used to explain the mechanism, not an additional benchmark result.

G.2 Evaluated GeriMedBench Trace

Table G2 shows a released consultation trace with the observed evidence, ATLAS action, and decision effect aligned by stage. The alignment makes it possible to verify that each revision follows the information actually revealed at that point.

Table G2 follows one released GeriMedBench case from its visible state through three questions to the final medication report. The case exposes both a success and an inefficiency:

the first two questions retrieve the decisive renal and gastrointestinal facts, whereas the third question adds no new risk information.

The trace shows why GeriMedBench separates information acquisition from final-decision quality. ATLAS obtains both critical facts and returns the complete reference-aligned decision, yet the third question does not improve the plan. The case contributes positively to Final Strict while leaving room for better Query Efficiency.

G.3 Why One-Shot Accuracy Is Insufficient

The evaluated traces illustrate why the final answer alone cannot reveal whether a system asked an informative question, updated the state correctly, or revised the affected medication component for the right reason.

The case separates answer plausibility from decision validity. Ibuprofen may appear reasonable when the initial message is treated as a complete patient profile. The renal condition changes the decision. ATLAS succeeds only if it identifies the missing dependency, asks a focused question, updates the graph, revises the affected medication components, and preserves a trace from evidence to action.

G.4 Error-Analysis Template

Table G3 standardizes the evidence recorded for each error category. Using the same reviewer question and remediation target helps distinguish state-grounding failures from PMCG, risk, revision, evidence, and schema failures.

This section separates the clinician-review protocol from its aggregate outcomes. Table H1 defines the review dimensions, and Tables H2 and H3 report overall and benchmark-stratified ratings.

H.1 Review Design

Table H1 defines the blinded review dimensions before any scores are summarized. This definition separates clinical correctness and safety from completeness, actionability, evidence consistency, and overall preference.

Three clinicians independently reviewed 40 benchmark-stratified cases, with 20 cases from the Western evaluation

Method	Strict Success Rate	OSRS
GPT-5	76.47	94.77
Claude Opus 4.6	81.21	95.18
Mistral-Small-3.2-24B	53.27	91.21
Qwen3-30B-A3B	79.41	95.82
DeepSeek-R1-Distill-Qwen-32B	77.94	96.03
MedGemma-27B-Text	78.59	96.36
Gemini 3.1 Pro Preview	92.60	97.84
Llama-3.3-70B-Instruct	56.05	90.73
RotatE	0.00	30.69
MDAgents	64.38	83.20
ATLAS	94.12	97.55

TABLE F3

NUMERICAL VALUES CORRESPONDING TO FIGURE 6 IN THE MAIN PAPER. STRICT SUCCESS IS A PERCENTAGE; OSRS USES A 0–100 SCALE.

Variant	Strict	M_{avoid} Recall	M_{caution} F1	Unsafe Rate	OSRS	Primary effect
Full ATLAS	92.04	100.00	75.47	0.00	97.21	Reference system
w/o PMCG	19.90	48.26	40.30	51.74	48.78	Removes patient-specific graph personalization and sharply degrades both safety and completeness.
w/o Geriatric Risk Auditor	38.81	93.53	40.30	6.47	80.47	Weakens caution and monitoring decisions.
w/o Drug Conflict Auditor	69.15	69.15	59.70	30.85	74.41	Misses medication–condition conflicts and avoidance decisions.
w/o Safety Gate	26.87	73.13	40.30	26.87	65.90	Allows unresolved or inconsistent decisions to reach the final output.

TABLE F4

KEY ABLATION EFFECTS REPORTED IN THE MAIN-TEXT ANALYSIS.

Step	Observed evidence	PMCG state	Decision consequence
Initial request	Older adult asks what to take for chronic knee pain. Hypertension is known. Kidney disease is unreported.	NSAID-related renal-risk dependency remains unresolved.	A one-shot recommendation would be premature.
Targeted question	ATLAS asks about kidney disease or renal function because the answer may change the safety of NSAIDs.	The unresolved renal-risk relation receives highest priority.	The system delays the final plan.
New evidence	The response reveals chronic kidney disease.	CKD activates a medication–condition conflict for NSAIDs.	NSAIDs move to M_{avoid} .
Alternative search	Guideline evidence supports acetaminophen within an appropriate use context.	Alternative relation links the excluded option to a safer candidate.	Acetaminophen enters M_{rec} , while a distinct safer strategy or guideline-supported substitute enters M_{alt} when appropriate.
Caution and monitoring	Blood pressure, pain response, dose, and follow-up remain clinically relevant.	Monitoring dependencies attach to the final plan.	M_{caution} records monitoring and use conditions.
Verification	Trace Verifier links the avoidance and alternative claims to CKD and guideline evidence.	Safety Gate finds no unresolved conflict.	ATLAS returns the structured plan.

TABLE G1

ILLUSTRATIVE PMCG UPDATE AND DECISION REVISION FOR THE KNEE-PAIN EXAMPLE.

Example G1: Presentation of structured decision-making and evidence pathways.

M_{rec} : Acetaminophen within an appropriate use context.

M_{avoid} : NSAIDs, including ibuprofen, because CKD increases kidney-injury risk.

M_{caution} : Monitor dose, blood pressure, pain response, and follow-up needs.

M_{alt} : Non-pharmacologic pain management or another guideline-supported option when appropriate.

U : No unsafe recommendation.

Evidence path: $\text{CKD} \rightarrow \text{renal-risk conflict} \rightarrow \text{NSAID avoidance} \rightarrow \text{safer alternative search}$.

and 20 from GeriMedBench. Each reviewer compared one ATLAS output with one Gemini 3.1 Pro Preview output.

We randomized case order and A/B assignment for each reviewer, removed model identifiers and automated scores, and

Stage	Evidence available	ATLAS action	Decision effect
Initial state	Age 79; analgesia need; aspirin 325 mg or nonselective NSAIDs under review; acetaminophen and nondrug support are visible candidates.	Marks renal and gastrointestinal safety dependencies as unresolved.	Delays the final plan rather than treating the initial profile as complete.
Query 1: renal function	The environment reveals renal-risk context relevant to NSAID use and dose accumulation.	Updates the patient state and activates the renal-risk conflict.	Strengthens the avoidance case for aspirin and nonselective NSAIDs.
Query 2: gastrointestinal history	The environment reveals gastric or acid-peptic disease with increased ulcer and bleeding risk.	Adds the gastrointestinal conflict to the PMCG and re-audits the candidate set.	Confirms that the high-risk option should move to M_{avoid} .
Query 3: heart-failure status	No additional heart-failure decompensation is documented.	Closes a remaining safety branch without activating a new conflict.	No medication component changes.
Final report	The two critical facts are available and linked to the medication decision.	Recommends acetaminophen, avoids aspirin or nonselective NSAIDs, returns a safe alternative, and passes the Safety Gate.	The case satisfies the complete structured reference under the evaluator.

TABLE G2

RELEASED GERIMEDBENCH TRACE FOR A 79-YEAR-OLD PATIENT SEEKING ANALGESIA. THE CASE IS GUIDELINE-DERIVED AND CONTAINS NO IDENTIFIABLE PATIENT INFORMATION.

Error category	Reviewer question	Recommended evidence to inspect
Missed information	Did the system leave a decision-changing fact unresolved?	Question ranking and unresolved PMCG relations.
Patient-state error	Did the system parse the answer into the correct state field?	Grounded state before and after the response.
PMCG error	Did the new fact activate or resolve the expected relation?	Graph transition record.
Risk-classification error	Was the fact assigned to avoidance, caution, or monitoring correctly?	Auditor output and guideline relation.
Alternative error	Did the substitute remain safe under the updated patient state?	Alternative relation and revised safety audit.
Revision error	Did every affected component change after the new evidence?	Before/after structured decision diff.
Evidence-alignment error	Does each final claim follow from a visible or acquired fact?	Evidence path and trace-verification output.
Schema error	Did the final output satisfy the required structured format?	Parser report and missing-field record.

TABLE G3

TRACE-ORIENTED TEMPLATE FOR QUALITATIVE ERROR ANALYSIS.

presented both outputs through the same structured template.

H.2 Overall and Benchmark-Stratified Results

Table H2 reports the aggregate clinician ratings and unsafe-case judgments. Table H3 separates the Western and GeriMedBench subsets to show whether the overall pattern is concentrated in one benchmark.

ATLAS receives higher mean ratings in all five criteria in both benchmark strata. The separation is larger on the Western cases, where the automated evaluation also shows stronger complete-decision performance. The reviewed sample remains small, and the results do not establish real-world clinical effectiveness.

H.3 Agreement and Interpretation

The agreement analysis should be read together with Tables H2 and H3. The clinician study is a blinded assessment of the sampled outputs and is not evidence of prospective clinical effectiveness.

Ordinal Krippendorff’s alpha ranges from 0.33 to 0.71 across the five criteria. The variation indicates that agreement depends on the criterion and should be interpreted with the small sample and ordinal rating scale in mind.

This section reproduces the shared task-facing instructions needed to interpret parser behavior and invalid outputs. Prompt I1 covers static decisions, Prompt I2 defines interactive actions, and Schema I1 gives the common structured medication-safety output.

I.1 Shared Static-Baseline Prompt

Prompt I1 shows the common static task instruction, including the visible patient state, candidate medications, and required structured decision fields.

I.2 GeriMedBench Action Format

Prompt I2 restricts each interactive response to either one question or one final structured decision, which keeps environment transitions machine-checkable.

Review item	Definition
Clinical correctness	Whether the medication decision is clinically appropriate for the presented patient state.
Medication safety	Whether the plan avoids harmful options and states necessary safety constraints.
Decision completeness	Whether the output covers recommendation, avoidance, caution, and alternative needs.
Actionability	Whether the plan gives a usable next step, monitoring action, or follow-up instruction.
Evidence consistency	Whether the rationale matches the patient evidence and guideline-grounded decision.
Potentially unsafe	Case-level judgment that an output contains a potentially unsafe recommendation.
Preferred output	Reviewer preference between the two blinded outputs, with ties allowed.

TABLE H1
CLINICIAN-REVIEW DIMENSIONS.

Method	Correct	Safety	Complete	Action	Evidence	Unsafe	Preferred
Gemini 3.1 Pro Preview	3.77	4.12	3.67	3.85	3.80	2/40	5/40
ATLAS	4.11	4.35	4.01	4.01	4.08	1/40	28/40

TABLE H2
OVERALL BLINDED CLINICIAN EVALUATION. SCORES USE A FIVE-POINT SCALE AND AVERAGE ALL REVIEWER–CASE RATINGS. UNSAFE COUNTS USE CASE-LEVEL MAJORITY JUDGMENTS. THE REMAINING 7 CASES WERE RATED AS TIES.

Benchmark	Method	Correct	Safety	Complete	Action	Evidence
Western	Gemini 3.1 Pro Preview	3.97	4.30	3.90	4.02	4.02
Western	ATLAS	4.45	4.60	4.40	4.22	4.37
GeriMedBench	Gemini 3.1 Pro Preview	3.57	3.95	3.43	3.68	3.58
GeriMedBench	ATLAS	3.77	4.10	3.62	3.80	3.78

TABLE H3
BENCHMARK-STRATIFIED MEAN CLINICIAN RATINGS.

Prompt I1: Shared static-baseline instruction template.

System role: You are a medication-safety decision-support model for older adults.

Input:

- Patient state: {visible_patient_state}
- Candidate medications: {candidate_set}
- Task: produce a complete structured medication-safety decision.

Requirements:

1. Recommend only supported options.
2. Identify medications that should be avoided.
3. State cautions, monitoring, dose, or duration conditions.
4. Provide safer alternatives when an option is excluded.
5. Do not assume that missing information is absent.
6. Return valid JSON only and follow the required schema.
7. Do not mention or infer evaluation labels.

Prompt I2: Interactive action contract for GeriMedBench.

At each turn, return exactly one action:

ASK:

```
{
  "action": "ask",
  "question": "<one safety-critical question>"
}
```

or FINAL:

```
{
  "action": "final",
  "decision": { ... structured medication decision ... }
}
```

You may ask at most $K = 3$ questions. Base each question only on the visible state and previously revealed answers.

1.3 Structured Decision Schema

parser and evaluator.

Schema I1 defines the medication sets, monitoring conditions, unsafe flag, and evidence paths consumed by the shared

Schema I1: The structured medication-safety output.

```
{
  "M_rec": [
    {"medication": "...", "use_context": "...", "evidence_path": ["..."]}
  ],
  "M_avoid": [
    {"medication": "...", "reason": "...", "evidence_path": ["..."]}
  ],
  "M_caution": [
    {"medication": "...", "monitoring_or_condition": "...", "evidence_path": ["..."]}
  ],
  "M_alt": [
    {"medication_or_strategy": "...", "rationale": "...", "evidence_path": ["..."]}
  ],
  "M_level": "...",
  "U": false,
  "sources": ["..."]
}
```

1.4 Parsing and Invalid-Output Handling

The invalid-output policy applies the same structural requirements described in Schema I1. A response is scored only after parsing and normalization produce the expected fields without introducing information absent from the model output.

The shared parser extracts the first valid JSON object, normalizes equivalent field names, and checks the required components. It does not fill missing clinical content from free text. Outputs that remain invalid after the shared runner policy receive the common failure treatment defined by the evaluation protocol.